



MD Mominul Islam Pramanik

Senior Application Engineer
AMI · HES · MDM · Full-Stack

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github.com/mominulIslam135/mis-cv

SKILLS

LANGUAGES

JavaScript · TypeScript · PHP · Dart · SQL · PL/SQL · C

BACKEND

Node.js · Express 4/5 · Sequelize ORM · REST APIs · Socket.io · JWT · Zod · Bcrypt · Winston · Resend · Jest

FRONTEND

Vue 3 (Composition API) · Vite · Pinia · Chart.js · Axios · Tailwind · Bootstrap 5 · Heroicons · vue-i18n

MOBILE

Flutter · Dart · SQLite · Dio · Provider · Firebase FCM · Google ML Kit · Flutter Map

DATABASES

Oracle DB (CISADM) · MS SQL Server · PostgreSQL · MySQL · SQLite

UTILITY DOMAIN

AMI · HES · Oracle MDM · Oracle CC&B · C2M · Smart Metering · Utility Billing · RC/DC · NOCS · OTP/SMS Gateway · BREB · DPDC

DEVOPS

CI/CD · PM2 · Linux · Cron Scheduling · Git · Deployment · Log Management

INTEGRATIONS & REPORTS

Telegram Bot API · MS Teams Webhooks · Email Automation · ExcelJS · PDFKit · OCR · PDF Parsing

AI & TOOLS

Claude · ChatGPT · Gemini · Prompt Engineering · AI Workflow Automation

EDUCATION

BSc in Computer Science & Engineering

Daffodil International University

2017 – 2021

LANGUAGES

Bengali — Native

English — Professional

PROFILE

Senior Application Engineer with 4+ years of experience building enterprise software for the utility and energy sector. Expert across the full Advanced Metering Infrastructure (AMI) stack — Head-End Systems (HES), Oracle Meter Data Management (MDM), and C2M billing operations. Architects, builds, deploys, and maintains production systems serving 2.6 lakh+ (260,000+) customers across DPDC and BREB AMI projects. Work spans full-stack web, cross-platform mobile, automation bots, CI/CD pipelines, system architecture, and AI-assisted development workflows. Proven track record leading stack migrations and collaborating with internal and external utility stakeholders.

EXPERIENCE

Senior Engineer, Application

Dec 2025 – Present

Oculin Tech BD Ltd. · Dhaka, Bangladesh

Leads full-stack architecture and development for AMI platforms serving 2.6 lakh+ customers across DPDC and BREB.

- Architects and ships **DPDC AMI Dashboard** — Node.js + Express + Vue 3 + PostgreSQL + Oracle CISADM. Live RC/DC monitoring, NOCS management, bank-wise payment reconciliation, an automated Missing Reads Finder, and 3-tier data recovery with local SQLite meter store.
- Architects and ships **OTBL CMS** — Vue 3 + TypeScript + Node.js + Sequelize + SQLite. Jira-style ticketing (epics, sprints, burndown), 5-level RBAC, 300k+ row Excel imports, CMO API integration with external SQL Server.
- Builds **CMO Mobile App** — Flutter offline-first with ML Kit OCR meter scanning, GPS/map, Firebase FCM push, self-hosted APK updates (currently v1.5.4).
- Leads the **BREB Smart Meter customer portal** (Flutter app + web, OTP auth, Oracle CC&B integration) and the **BREB CMO → MDMS migration platform** (offline-first Flutter field app, real-time Vue 3 admin-web).
- Designs and ships REST APIs across every platform — CMO batch sync, BREB customer/admin APIs, and CMS integrations — covering auth, file upload, reporting, and real-time (Socket.io) updates.
- Designs automation: Telegram bots (live RC/DC stream, OTP-secured customer bot), MS Teams hourly batch reports with stuck-batch alerts, scheduled daily email summaries.
- Sets up CI/CD pipelines, PM2 process management, Linux deployments, scheduled cron jobs (hourly NOCS refresh, daily 2 AM bill stop analysis, 30-min batch monitoring).
- Operates Oracle MDM and C2M for batch processing, billing reconciliation, and day-to-day operational problem-solving. Coordinates with internal ops and external DPDC/BREB stakeholders.
- Leads system design and architecture across multi-DB, multi-platform stacks. Applies AI tools (Claude, ChatGPT, Gemini) for design review, code review, and workflow automation.

Stack: Node.js · Express · Vue 3 · TypeScript · Flutter · Oracle · MS SQL · PostgreSQL · SQLite · Sequelize · Socket.io · PM2 · Linux · Telegram Bot API

Junior Engineer, Application

Feb 2023 – Dec 2025

Oculin Tech BD Ltd. · Gulshan, Dhaka

- Built dashboards and reporting tools using PHP, Vue.js, JavaScript, MySQL, Bootstrap for utility operations.
- Worked on AMI data ingestion, HES integration, Oracle MDM batch operations, and C2M billing workflows.
- Automated CSV/Excel/PDF report generation for billing engine input and ops team consumption.
- Supported live DPDC AMI rollouts and contributed to BREB AMI deployment for 2.6 lakh+ customers.
- Resolved production issues across DB, backend, and frontend layers — billing discrepancies, batch failures, meter data sync errors.
- Led team migration to modern stack — Node.js, Vue 3, TypeScript, Tailwind CSS, Vite.

Stack: PHP · Vue.js · JavaScript · MySQL · Oracle MDM · C2M · Bootstrap · HTML5/CSS

Application Team Intern

Jun 2022 – Jan 2023

Oculin Tech BD Ltd.

- Contributed to live utility software projects from day one — features in PHP, JavaScript, MySQL across internal tools.
- Worked on HES, Oracle MDM, and operational dashboards. Took ownership end-to-end — earned full-time offer.

KEY PROJECTS

DPDC AMI Dashboard — Full-Stack AMI Operations Platform

Production AMI platform serving DPDC ops teams. Features live RC/DC command tracking with NOCS area breakdown, pending IMD tracking, batch monitoring with auto-alerts, NOCS balance summaries and installation progress, bank-wise payment reconciliation, bill stop management with PDF data parsing, and customer billing history. Automated multi-channel reporting: Microsoft Teams (hourly batch operational reports, stuck-batch alerts), Telegram Ops Bot (live RC/DC command stream), OTP-secured customer Telegram bot, and scheduled email summaries to area managers. Implements 3-tier data recovery model with local SQLite meter store. Serves 2.6 lakh+ customers.

Node.js · Express 4 · Vue 3 (Composition API) · Vite · Tailwind · Pinia · PostgreSQL · Oracle CISADM · SQLite · Sequelize · Winston · JWT · Bcrypt · Resend · ExcelJS · PDFKit · PM2 · Telegram Bot API

OTBL CMS — Utility Management Platform

Internal full-stack platform unifying operations for a power distribution utility. Jira-style ticket management (epics, sprints, subtasks, audit trail); analytics dashboard (burndown, sprint velocity); complaint management with priority handling; bill stop analysis to prevent revenue leakage; meter replacement and estimator tracker; automated Read Upload for Bill Start (CSV generation); batch reports with error detection; bulk customer upload (300k+ rows); CMO API integration with external SQL Server. 5-level RBAC (Super Admin → User).

Vue 3 · TypeScript · Vite · Tailwind · Bootstrap 5 · Node.js · Express 5 · Sequelize · SQLite · JWT · Telegram Bot · AI Integration

CMO Mobile App — Field Operations (Android v1.5.4)

Cross-platform Flutter app for field technicians managing meter operations. Full offline-first architecture with local SQLite, Google ML Kit OCR for meter number scanning, Flutter Map for GPS field navigation, Firebase FCM push notifications, role-based access (Admin / Field User), CSV export with filters, and self-hosted APK update system. Backend on Node.js + Microsoft SQL Server with JWT auth.

Flutter · Dart · SQLite · Google ML Kit · Firebase FCM · Dio · Provider · Flutter Map · Node.js · Express · MS SQL Server · JWT

BREB Smart Meter — Customer Self-Service Portal (App + Web)

OTP-authenticated self-service portal for BREB electricity customers — one Flutter codebase shipped to Android and web. Customers track live meter usage (today/week/month kWh with hourly breakdown), day-wise and monthly billing, and recharge

history across multiple linked accounts, with dark mode and Bangla/English support. Backend reads billing data directly from the Oracle CC&B/C2M database (strictly read-only) with PostgreSQL for OTP and account state, and Banglalink SMS for OTP delivery. Paired with a Vue 3 ops dashboard for real-time RC/DC monitoring, PBS-scoped role-based access, and payment/recharge/tariff reporting.

[Flutter](#) · [Dio](#) · [go_router](#) · [Node.js](#) · [Express 4](#) · [Oracle CC&B](#) · [PostgreSQL](#) · [JWT](#) · [Banglalink SMS API](#) · [Vue 3](#) · [Pinia](#) · [Tailwind](#)

BREB CMO → MDMS Migration Platform (Admin Portal + Mobile App)

End-to-end platform migrating BREB's legacy Change-Meter-Owner (CMO) records into Oracle MDMS. A Flutter field app captures meter installations with ML Kit OCR reading, barcode/QR scanning, GPS geotagging, and full offline-first customer search across a 70+ field schema, syncing once back online. A Vue 3 + TypeScript admin-web (real-time via Socket.io, Chart.js analytics, Bangla/English i18n) tracks migration records and approvals. Backend on Node.js/TypeScript + Express + PostgreSQL with JWT auth and an automated Jest test suite.

[Flutter](#) · [Google ML Kit](#) · [SQLite](#) · [Node.js](#) · [TypeScript](#) · [Express](#) · [Sequelize](#) · [PostgreSQL](#) · [Socket.io](#) · [Vue 3](#) · [Chart.js](#) · [JWT](#) · [Jest](#)